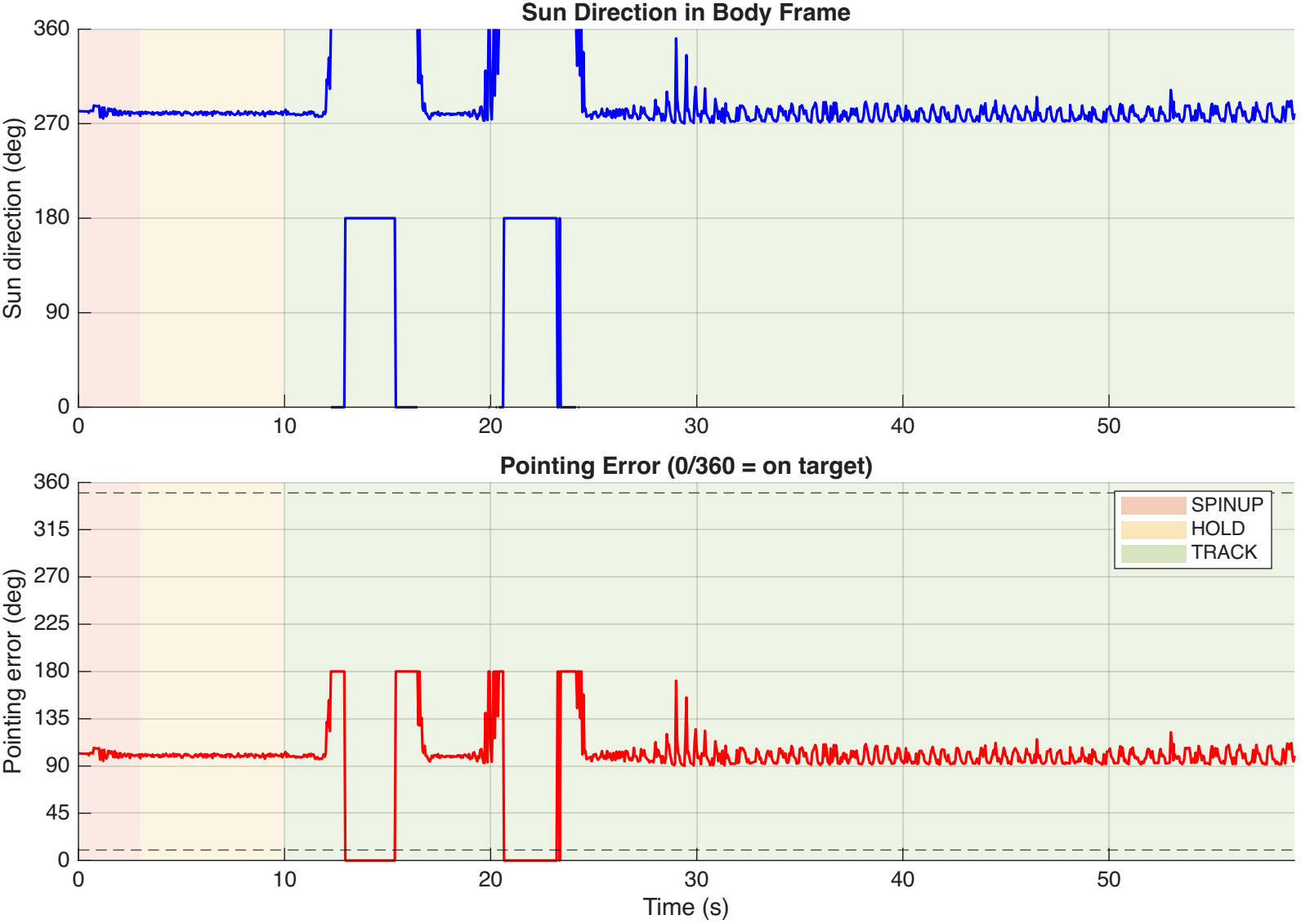
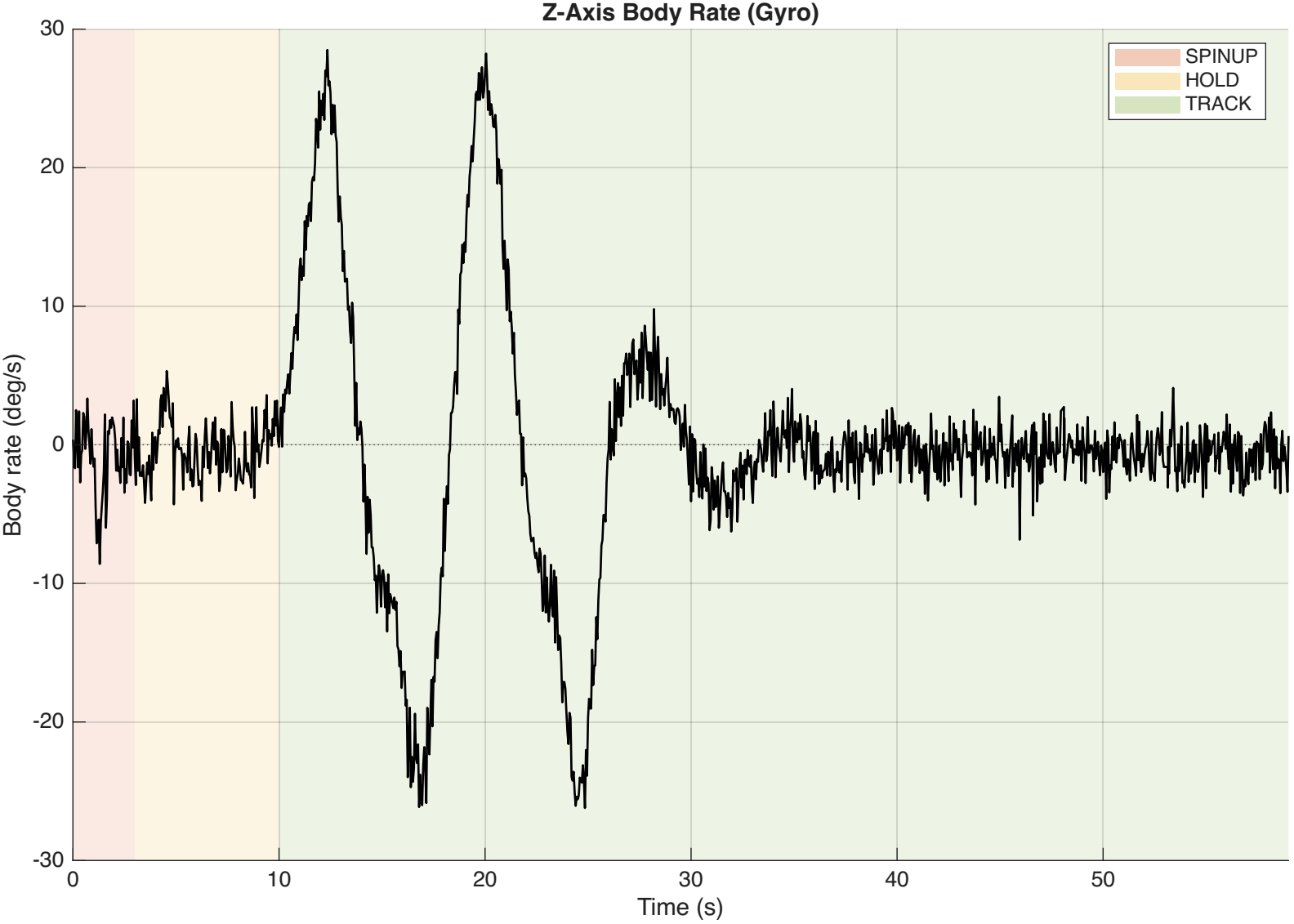
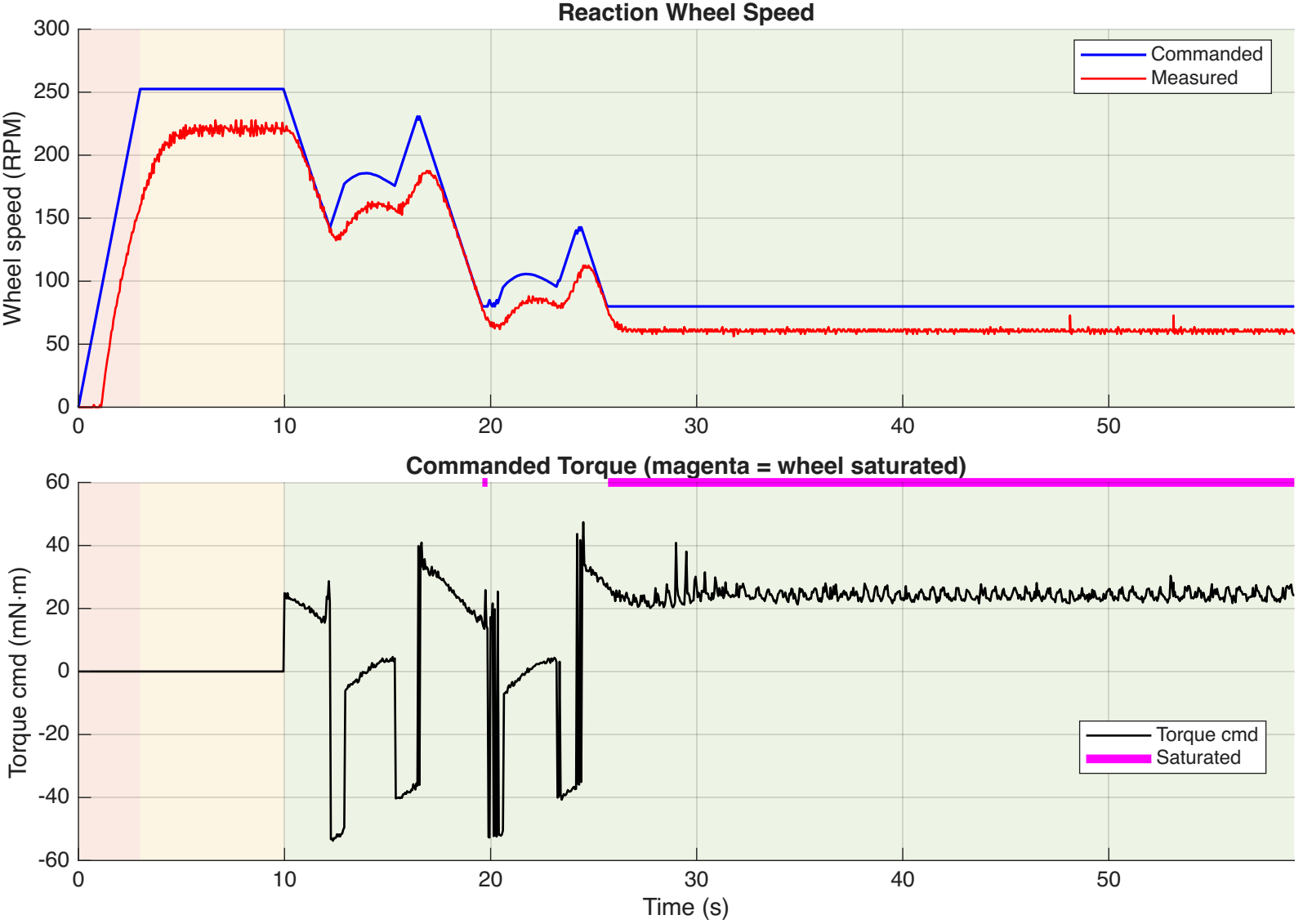
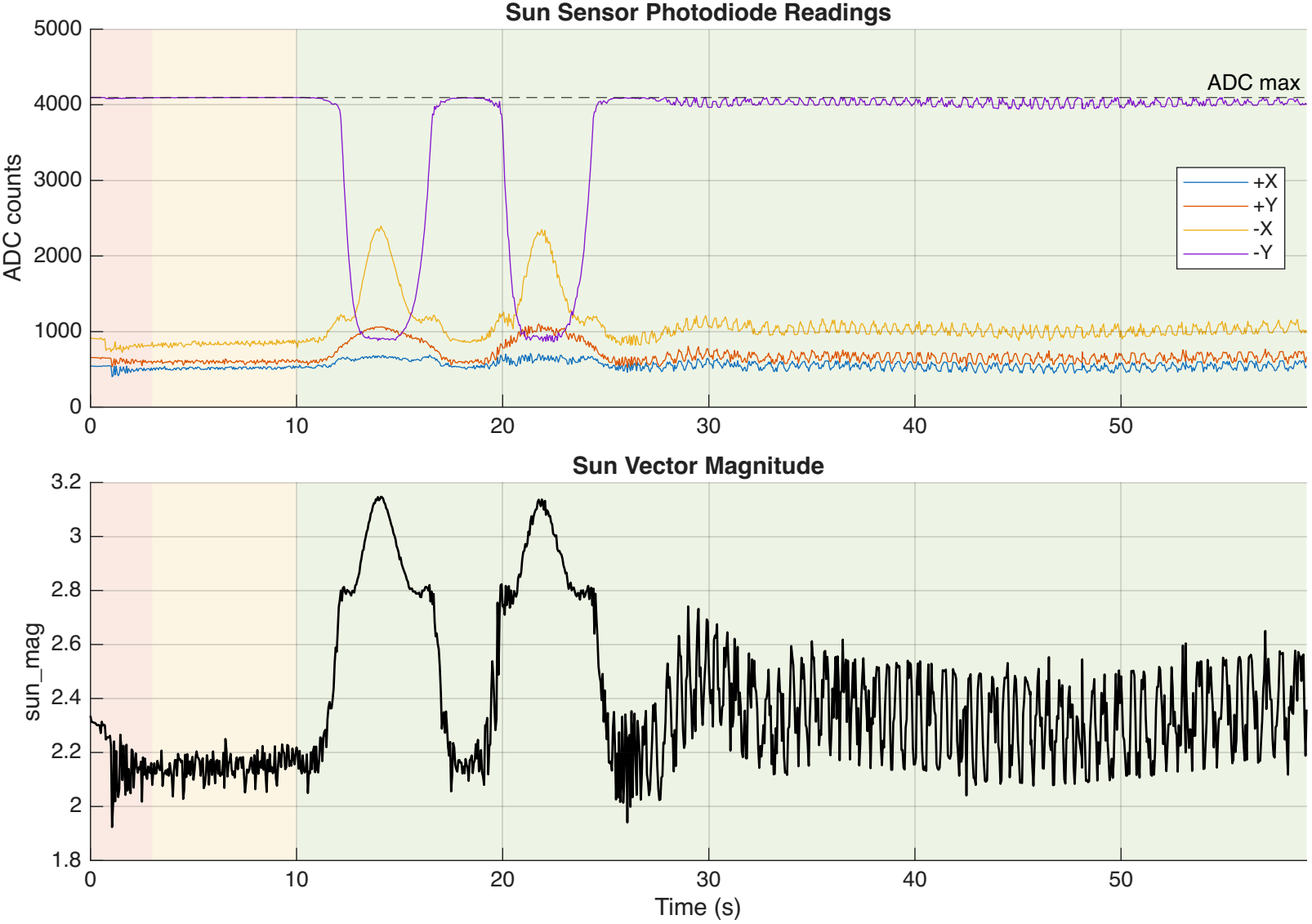

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
10
20
30
40
50
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)


Z-Axis Body Rate (Gyro)
SPINUP
HOLD
TRACK
Body rate (deg/s)
30
20
10
0
-10
-20
-30
0
10
20
30
40
50
Time (s)


Reaction Wheel Speed
Wheel speed (RPM)
Commanded
Measured
300
250
200
150
100
50
0
0
10
20
30
40
50
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
Torque cmd
Saturated
60
40
20
0
-20
-40
-60
0
10
20
30
40
50
Time (s)


Sun Sensor Photodiode Readings
ADC counts
5000
4000
3000
2000
1000
0
ADC max
+X
+Y
-X
-Y
0
10
20
30
40
50
Sun Vector Magnitude
sun_mag
3.2
3
2.8
2.6
2.4
2.2
2
1.8
Time (s)